

Mathematical Foundations of AI & ML

Philipp Pelz

1 Mathematical Foundations of AI & ML - proposed syllabus

Week	MFML – Mathematical Foundations	ML-PC – ML in Materials Processing & Characterization	MG – Materials Genomics	Exercise (90 min, Python-based)
1	Data as vectors, models, feature spaces	What counts as data in processing & characterization	What counts as data in materials databases	NumPy arrays, vector operations, plotting simple datasets
2	Data types, scales, normalization, units	Sensors, signals, images as arrays	Descriptors, compositions, metadata	Scaling & normalization effects; visual comparison
3	Probability, expectation, variance, noise	Measurement noise in experiments	Noise & uncertainty in databases	Monte Carlo sampling, simulate noisy measurements
4	Sampling, Nyquist, FFT as representation change	Signal preprocessing (spectra, time series)	Property distributions & statistics	FFT on synthetic signals; filtering
5	Linear regression, least squares (geometry)	Regression for process/property modeling	Correlation analysis in materials data	Implement linear regression from scratch
6	SVD & PCA (eigenvectors, variance)	PCA on spectra & micrographs	PCA & embeddings for materials spaces	PCA on high-dimensional dataset, visualize components
7	Loss functions, gradients, sensitivity	Model tuning & objective functions	Descriptor relevance & feature selection	Manual gradient descent on toy problems
8	Optimization, regularization (L1/L2)	Model robustness & stability	High-dimensional regression challenges	Compare regularization strengths
9	Neurons, activations, backpropagation	NN regression/classification for properties	NN for structure–property mapping	Tiny neural net: forward + backward pass